



30 Years of Advancement and Challenges in Early HIV Diagnosis in Jamaica

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Background

The first case of HIV in Jamaica was reported in 1982 and antiretrovirals (ARV) became available at public treatment sites in 2004. This study examines the epidemiology of HIV among persons living with HIV (PLHIV) from 1990 to 2020.

Methods

We analysed data from PLHIV first enrolled at HIV treatment sites in Jamaica between January 1990 to December 2020 to identify changes in PLHIV demographics, clinical stage, and location. The data span three periods: 1990-2003 (P1) pre-ARV access; 2004-2016 (P2) - Public ARV for PLHIV with CD4 <350; and 2017-2020 (P3) - Universal Test and Treat in Jamaica. We used chi-square and Kruskal-Wallis H tests to analyze changes across the periods.

Results

25,893 PLHIV registered at treatment sites: 731 were in P1, 19,479 in P2, and 5,683 in P3, Significant differences emerged in sex distribution ($\chi^2 (2) = 42.37, p < 0.001$), age ($\chi^2 2 (6) = 1142.42, p < 0.001$), area of residence ($\chi^2 (4) = 132.89, p < .001$), and WHO HIV stage at registration ($\chi^2 (6) = 368.70, p < 0.001$).

Table 1. Demographic and clinical characteristics of male and female PLHIV receiving treatment in Jamaica from 1990 to 2020. (n= 25,893)

		1990 - 2003		2004 - 2016		2017 - 2020		P-value*
		n	%	n	%	n	%	
Sex at Birth	Total	731		19479		5683		
	Male	339	46.4%	9152	47.0%	2946	51.8%	<.001
	Female	392	53.6%	10327	53.0%	2737	48.2%	
Age at Group at Registration	Median Age at Registration (IQR)	29 (Q1=16, Q3=36)		35 (Q1=27, Q3=45)		36 (Q1=25, Q3=48)		
	< 15	165	22.6%	643	3.3%	89	1.6%	<.001
	15 - 24	106	14.5%	2985	15.3%	1169	20.6%	
	25 - 49	426	58.3%	12675	65.1%	3162	55.6%	
	50 +	32	4.4%	3081	15.8%	1257	22.1%	
	Missing	2	0.3%	95	0.5%	6	0.1%	
WHO HIV Staging	Median Baseline CD4 Result cell/mm(IQR)	310 (Q1=118.5, Q3=533.8)		291 (Q1=103, Q3=503)		389 (Q1=198, Q3=608)		<.001
	Stage 1	186	25.4%	4684	24.0%	1835	32.3%	
	Stage 2	112	15.3%	3138	16.1%	975	17.2%	
	Stage 3	122	16.7%	3595	18.5%	997	17.5%	
	Stage 4	236	32.3%	7059	36.2%	1280	22.5%	
	Missing	75	10.3%	1003	5.1%	596	10.5%	
Regional Health Authority	SERHA	575	78.7%	10087	51.8%	2887	50.8%	<.001
	WRHA	62	8.5%	4291	22.0%	1314	23.1%	
	SRHA	32	4.4%	2543	13.1%	731	12.9%	
	NERHA	62	8.5%	2558	13.1%	751	13.2%	
Parish of Residence at Registration	Urban	361	49.4%	6361	32.7%	2152	37.9%	<.001
	Tourist	79	10.8%	5112	26.2%	1523	26.8%	
	Rural	155	21.2%	4455	22.9%	1355	23.8%	
	Missing	136	18.6%	3551	18.2%	653	11.5%	

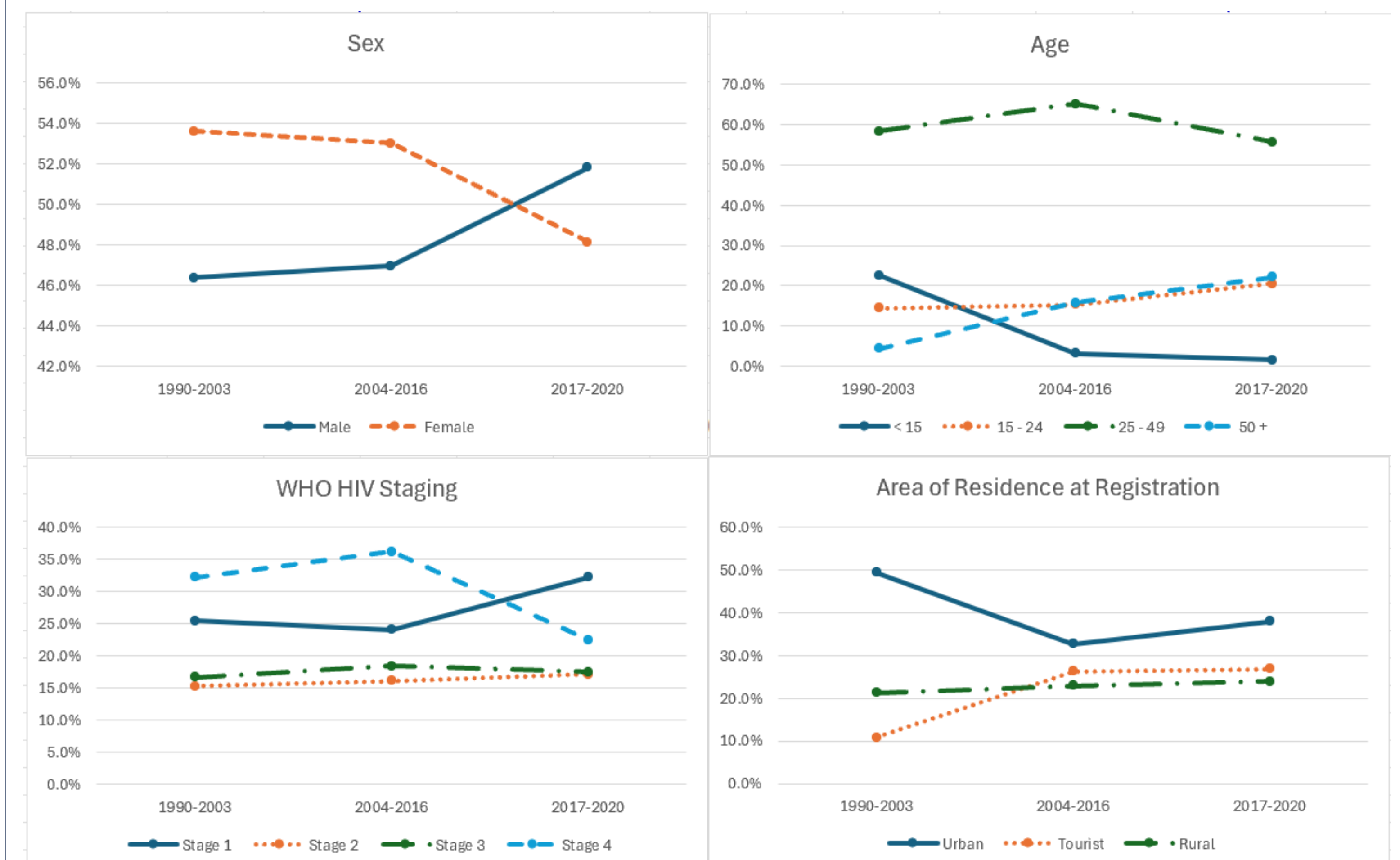


Figure 1. Changes in demographics of PLHIV enrolling in care in Jamaica 1990-2020. (n=25,893)

Median age at registration increased from 29 in P1 to 36 in P3; ($H (2) = 260.59, p\text{-value} < 0.001$). Baseline CD4 counts increased from P1 to P3 ($H (2) = 424.26; p < .001$). Stage 4 diagnoses decreased in PLHIV ages under 15, 15-24, and 25-49, but not among PLHIV 50 years and older. Similarly, Stage 1 diagnosis increased across all age groups, except those in the 50 and over cohort. PLHIV residing in tourist areas increased while those from urban areas decreased ($\chi^2 (4) = 132.89, p < .001$).

Conclusion

The decline in pediatric HIV cases and overall increase in early-stage diagnoses are notable public health achievements. However, Jamaica has an aging cohort of PLHIV entering treatment and targeted testing strategies and behavioural research are needed to reduce HIV transmission and late-stage diagnoses in older adults

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